

VectorShift Workflow Studio

Comprehensive Features & User Documentation Guide

1. Multi-Workflow Sheet Tabs (+ New Workflows & Rename)

- **Create New Workflows (+ Button):** Click the '+' icon on the bottom tab bar to create new workflow tabs with sequential auto-naming (e.g., Untitled 1, Untitled 2).
- **Tab Switching & Navigation:** Click any workflow tab at the bottom to switch instantly between active workflows while preserving full canvas state.
- **Inline Tab Renaming:** Double-click any tab name to rename it inline. Press Enter or click outside to save the new name.

2. Excel-Style Save or Don't Save Modal Prompt

- **Unsaved Tab Close Handler:** When clicking the 'X' close button on an unsaved tab, an Excel-style confirmation modal appears asking to Save or Don't Save.
- **Save Action:** Triggers native laptop file save picker (.json) to save the workflow directly to your computer/desktop, marks the tab as saved, and closes cleanly.
- **Don't Save Action:** Discards unsaved changes and closes the tab immediately without popups or moving to trash.
- **Saved Tab 1-Click Close:** Saved tabs close cleanly with zero popups and zero clutter.

3. Open / Import Existing JSON Files (📁 Icon)

- **Import Existing Files:** Click the folder icon (📁) on the bottom tab bar to select and parse any JSON workflow file from your computer/laptop directly into active studio sheets.

4. Hand Pan Mode vs. Arrow Selection Mode

- **Hand Pan Mode (🖱️):** Click and drag anywhere on the canvas background to pan smoothly. Connect node handles by dragging lines between dots. Double-click any edge connection line to delete it.
- **Arrow Selection Mode (🔍):** Click and drag a box over multiple nodes to batch select, move, or copy. Double-click edge deletion is suppressed in Arrow mode to prevent accidental deletion during box selection.

5. Dual-Layer Laptop File Deletion & 30-Day Studio Trash

- **Laptop File Deletion Auto-Detection:** If a saved JSON file is deleted from your computer/desktop or recycle bin, VectorShift Studio automatically detects the missing file when the window regains focus.
- **30-Day Studio Trash Retention:** Missing or deleted workflows are safely preserved in Studio Trash for 30 days before auto-purging.
- **Trash Management Submenu:** Click Three-Dots Menu ➔ Trash to view deleted items, restore them back to active tabs, or permanently empty trash.

6. Copy, Paste & Undo/Redo Engine

- **Copy & Paste (Ctrl+C / Ctrl+V):** Select nodes and press Ctrl+C (Cmd+C) to copy. Press Ctrl+V (Cmd+V) to paste copied nodes onto the canvas with offset positioning.
- **Undo & Redo (Ctrl+Z / Ctrl+Y):** Full 50-step state history. Press Ctrl+Z to undo and Ctrl+Y (or Ctrl+Shift+Z) to redo any action.

7. Deleting Nodes & Edges (Cross Button / Double Click)

- **Node Removal (Cross Button X):** Click the 'X' button on the top-right corner of any node card to delete it instantly.
- **Edge Removal (Double Click):** Double-click any connection line on the canvas to delete it instantly.

8. Appearance & Theme Engine (Default, Light, Dark)

- **Theme Submenu:** Click Three-Dots Menu → Appearance to toggle between Default (VectorShift Official Brand), Light Mode, and Dark Mode with smooth CSS transitions.

9. Touchpad & Touch Zoom Engine

- **2-Finger Touchpad Zoom:** Slide 2 fingers up/down on your laptop touchpad to zoom in and out smoothly without needing Ctrl.
- **Touchpad Pinch Zoom:** Pinch in/out with 2 fingers on trackpad to zoom in/out.
- **Ctrl + Scroll Wheel:** Holding Ctrl + mouse wheel scroll continues to work as expected.

10. Real Pipeline Execution Engine (Bonus Feature)

- **Analyze Pipeline (/pipelines/parse):** Calculates node count, edge count, and validates DAG cycle status.
- **Run Execution Engine (/pipelines/execute):** Executes nodes topologically: substitutes {{variables}}, evaluates Math/Filter logic, awaits Timer delays, calls real/mock LLM API, and displays step-by-step execution logs and terminal outputs.